logisth 0.5.0 — User Manual



A two-part polyphonic subtractive synthesizer for VST3, CLAP and standalone use. This manual describes logisth's own controls and behaviour. logisth is an independent instrument and is not affiliated with LennarDigital. It does not load Sylenth1 presets or reproduce its sound exactly.

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1. Installation and first sound

Download the archive for your operating system from the GitHub release. Each native archive contains VST3, CLAP and Standalone folders. Extract it first. Copy the entire .vst3 bundle and the .clap file or bundle into a plug-in directory scanned by your DAW. Keep the internal structure of bundles intact.

System	User VST3 directory	User CLAP directory
Windows	%LOCALAPPDATA%\Programs\Common\VST3	%LOCALAPPDATA%\Programs\Common\CLAP
macOS	~/Library/Audio/Plug-Ins/VST3	~/Library/Audio/Plug-Ins/CLAP
Linux	~/.vst3	~/.clap

If your DAW does not scan these user directories, add them to its scan paths or use its documented system directory. Rescan plug-ins, insert logisth on a MIDI instrument track, select Init / analog from SOUND LIBRARY, and play a note. The on-screen keyboard also generates notes. Start at a moderate listening level.

For Standalone, launch the executable or app and choose audio and MIDI devices in its audio settings. The standalone wrapper provides its own device settings; those are separate from the synth panel.

Windows builds target x64. macOS Universal bundles contain Intel x86_64 and Apple Silicon arm64 code and require macOS 11 or later. The minimum Linux glibc version is recorded in BUILD_INFO.json. macOS bundles have an ad-hoc signature, without Apple notarization; Windows binaries have no publisher signature. Follow your OS's normal review/open procedure for downloaded applications you trust.

VST2 is not included: building it requires a separately licensed VST2 SDK.

2. The single-window console

All sections occupy one console. No page needs vertical scrolling. Select A or B at the top to edit that part's two oscillators, filter and amplitude envelope. This selects the editing view; it does not mute the other part. Use SOLO for listening to one part. The central rack changes its controls when you select a rack name. The small square beside each name independently enables that section. Selecting a rack page does not enable or disable its processing.

The upper row contains the preset selector, previous/next preset buttons, A/B, SOLO, POLY, SYNC and THEME. Filters are on the left, the rack in the centre, and the mixer on the right. Mod Env, LFO and Misc routing remain visible below them. The bottom row contains wheels, the keyboard and mono/portamento controls.

Drag the bottom-right resize handle to scale the console from 80% to 150%. The nominal size is 1240 by 860 pixels. Initial size adapts to the display. THEME switches between LIGHT and DARK. Each instance saves its own theme with the plug-in state. The theme changes appearance without changing audio parameters.

Drag knobs and faders to adjust values. Numeric fields can be dragged or edited. Examples of accepted values include 2.5k for 2500 Hz, 350ms and 65%. Double-click a control to restore its default; modulation amount controls return to zero. Hover for a tooltip and the parameter/value readout at the bottom. Disabled phase knobs become available when their oscillator's RETRIG is enabled.

3. Oscillators and amplitude envelopes

There are four oscillators: A1 and A2 feed source A; B1 and B2 feed source B. Each oscillator has a waveform menu, 0–8 unison voices, volume, octave, note, fine tuning, phase, detune, stereo width, pan, retrigger and invert.

Control	Purpose
Wave	Selects the oscillator waveform.
Voices	Sets the unison voice count. Zero silences this oscillator.

Control	Purpose
Volume	Sets its level before filtering and the part envelope.
Octave / Note / Fine	Coarse octave, semitone and cent tuning.
Phase / Retrig	Retrig starts notes at the selected phase; otherwise phase runs without this reset.
Detune	Spreads unison pitches. Increase gradually for a wider ensemble.
Stereo	Spreads unison voices across the stereo field.
Pan	Positions the oscillator output left or right.
Invert	Inverts oscillator polarity; it can affect cancellation when sources mix.

The amplitude envelope for each part has attack, decay, sustain and release. Attack rises from silence, decay approaches sustain, and release fades after note-off. Sustain is a level; the other three are times. The line above the faders illustrates the current envelope. Amp Env A and B remain independent.

3.1 Four source engines

Every oscillator slot A1/A2/B1/B2 has its own engine menu: Classic, Wavetable, Sampler or Granular. All four may use different engines and different audio files. The adjacent view menu selects VOICE, SOURCE, ENV, GRAINS, MOTION or TIME without opening another window. VOICE holds the shared tuning, level, unison, stereo and pan controls. SOURCE holds the engine-specific controls; ENV holds that oscillator's optional ADSR. The central Amp Env A/B remains the final envelope for its part.

Classic retains the original oscillator waveforms and retrigger behaviour. Wavetable scans up to 256 frames using POSITION. Frames are interpolated and precomputed spectral bands reduce aliasing at higher notes. WARP bends phase. Choose a frame size (256/512/1024/2048 samples) before IMPORT when loading a concatenated wavetable WAV. The factory table morphs between sine and harmonics. Wavetable extraction uses the file's left channel. Padding is used for files shorter than one frame; additional frames beyond 256 are ignored.

Sampler plays the imported audio at the selected ROOT MIDI note (default 60). START and END choose a region; LOOP repeats it with a short crossfade; REVERSE reverses playback. Without LOOP, playback ends at the selected boundary. Root sets the reference pitch, not automatic pitch detection. Stereo files retain both channels; mono files feed both channels. Unison and pitch bend still apply.

Granular offers two models. **Granulizer** is the default for new patches; **Cloud** retains the eight Hann-windowed grains, size, density, scan and spray controls from 0.4. Existing projects select Cloud automatically.

In Granulizer, GRAINS contains ATTACK, HOLD, GRAIN SPACE and WAVE SPACE. ATTACK sets both fade-in and fade-out, so total grain duration is twice ATTACK plus HOLD. GRAIN SPACE is the interval between grain starts relative to HOLD; WAVE SPACE advances the source by HOLD times that percentage at every start. With zero ATTACK and both spacings at 100%, playback reconstructs the sample at its original pitch. Increasing GRAIN SPACE slows traversal without changing pitch. WAVE SPACE can be negative: this reverses the order of grains while each grain still reads forwards. REVERSE separately reverses each grain's samples. Up to 64 grains overlap per unison voice; further starts are skipped if full.

MOTION contains alternating grain PAN, wave-spacing modulation DEPTH and SPEED, RANDOM source-position variation, TIME HOLD (freezes traversal), LOOP and REVERSE. TIME contains transient mode Off / Regions / Detect, transient HOLD, KEY TO, TEMPO SYNC and BEATS. Regions uses imported WAV cue points; Detect uses a local onset detector. A crossed marker aligns the grain and replaces its hold time. KEY TO selects Pitch, Percent, Step or Transient. Pitch follows played notes; the other modes keep source pitch and use keys relative to ROOT for position: +12 semitones selects 50% of the sample, advances 12 holds, or selects region 12, respectively. Negative position offsets clamp to the first position/region. TEMPO SYNC fits traversal to BEATS at the current tempo without repitching grains.

START sets the initial location; automation moves the active scanner. LOOP wraps over the sample to END; TIME HOLD keeps creating grains at the current location.

Click an oscillator's waveform strip or select SAMPLE in the rack for the larger waveform view. Choose A1/A2/B1/B2. It draws the actual stereo amplitude envelope, region markers, start/end and the most recent active voice's scanner, plus up to 16 active grain positions. Click to change START; wheel zooms around the cursor; Shift-wheel pans. The display is a view of playback, not a sample editor.

Enable OSC ENVELOPE in ENV to give a slot an additional attack, decay, sustain and release. A part's shared envelope still limits its final output. New Position A/B/A+B and Grain size A/B/A+B modulation destinations scan frames or sample locations and scale grain duration. Position changes are smoothed.

IMPORT accepts WAV, AIFF and FLAC at 8–96 kHz, up to 12 seconds per source. Alternatively, drop one file onto an oscillator panel: a Classic slot switches to Sampler, while the other engine selections are retained. Import decoding and table preparation run on a worker, and the footer reports completion or errors. RESET returns the slot's material to the factory glass sample/table. The waveform strip previews the loaded material. No original media path is needed on project recall: imported PCM data is compressed into the plug-in state, with a 64 MiB state-input limit. Save larger libraries separately; this is a single-sample source rather than a multisample instrument.

4. Filters and shared Filter Control

Each part has an input selector, type, slope, cutoff, resonance and drive. Input A uses A1+A2, B uses B1+B2, A+B uses both, and None disconnects the input. Consequently, a part's filter can process the other oscillator source. Its output still passes through its own amplitude envelope and Mix level.

Choose low-pass, high-pass, band-pass or bypass. Slope selects 12 or 24 dB per octave. Cutoff is the local frequency. Resonance emphasises the cutoff region; high resonance can self-oscillate while a note is active. Drive adds saturation.

The shared FILTER CONTROL affects both A and B while retaining their local settings. CUTOFF is an offset in semitones: +12 doubles both cutoff frequencies, -12 halves them. RESO adds a shared resonance offset. KEYTRACK ties cutoff to played pitch; at full tracking, an octave higher doubles cutoff. WARM changes the filter saturation characteristic. There is no third filter in this section. Modulation is combined with these settings before the effective cutoff is limited to a valid frequency for the current sample rate.

5. Modulation

Mod Env 1 and 2 each provide an ADSR envelope and two destination/amount pairs. LFO 1 and 2 also provide two pairs each. Misc 1 and Misc 2 each contain two source blocks, with two pairs per block. There are 16 routes in total. The source menu in a Misc block chooses the signal used by both routes in that block.

Choose a destination, then move its small amount knob away from zero. Amount ranges from -10 to +10; negative values reverse the modulation. None or amount zero makes a route inactive. Routes to the same destination are added together. The destination popup groups oscillators, filters and modulation/effects into three columns.

Destinations include Volume, Pitch, Phase and Pan for A, B or A+B; Cutoff and Resonance for A, B or A+B; Mix A, B or A+B; Phaser Frequency; LFO 1/2 Rate and Gain; and Distortion Amount. Modulating A+B affects both parts together.

Misc sources include Velocity, Keytrack, Aftertouch, Mod Wheel, Pitch Wheel, Amp Env A/B, Mod Env 1/2, LFO 1/2, Step Velocity and random sources. Polyphonic pressure takes precedence over channel pressure for the affected note. Random values are generated at note-on. Step Velocity follows the active arp step and returns to zero after its gate ends.

Each LFO has waveform, Rate, Gain, Offset and FREE. With SYNC enabled and FREE off, RATE is a rhythmic division at the host tempo. With FREE enabled it uses its Hz rate and does not restart on each note. Otherwise notes restart the LFO. Gain scales its output; Offset shifts it. The global fallback tempo is 120 BPM when no host tempo is available.

Modulation depth is specific to logisth. At amount +10 and source value +1, pitch adds 48 semitones, cutoff adds 120 semitones, phase adds one cycle, and pan adds one full side. Volume, Mix, Resonance, Gain and Distortion add one

linear unit. LFO Rate multiplies by 16 and Phaser Frequency by 1024. Smaller amounts give smaller changes; start low. Effects are global, so FX modulation uses the newest active voice. LFO feedback modulation uses the previous sample.

6. Chords and pattern arpeggiator

The CHORDS rack page and Pattern arp use a chord-programming and multi-lane workflow inspired by Cthulhu, with logisth's own interface and implementation. It is not a Cthulhu clone and does not import Cthulhu chord/pattern files. The chain is MIDI input Chords Arpeggiator Synth and optional MIDI output. Each processor has its own enable square in the rack.

Chord mapper

Choose CHORDS and enable its square. TRIGGER selects one of 128 MIDI notes. VOICING assigns Unison, Major, Minor, Maj 7, Min 7, Dominant 7, Sus 2, Sus 4, Power, Diminished or Augmented to that trigger. MAP ALL assigns the selected type to all triggers, transposed to each root. Eight note fields edit the stored voicing directly: drag or enter MIDI numbers 0–127; -1 removes a voice. Empty triggers pass their original note through. Shared notes in overlapping chords are reference-counted so releasing one trigger does not cut another chord short.

LEARN collects up to eight incoming MIDI notes for the selected trigger. Play your voicing, then choose DONE. Notes are collected silently during learning; DONE stores the first eight distinct pitches in ascending order. The complete chord map is saved with the project. Chords can play as blocks with arp disabled, or supply a pool of notes to the Pattern engine.

Eight-graph arpeggiator

Select ARPEGGIATOR, choose Pattern, enable GRAPHS and the rack enable square. The selected graph occupies one drawing area with 32 steps. The tabs are:

Graph	Meaning
NOTE	Rest, held-note ranks 1–8, Up, Down, Up/Down, Down/Up, UpAndDown, DownAndUp, FingeredBottom or FingeredTop.
RAND SEL	Random positive or negative rank offset, wrapping in the held-note pool.
OCTAVE	-8 to +8 octaves.
PITCH	-24 to +24 semitones, with seven pitch-degree enable buttons below.
VEL SCALE	0–200% of input velocity; zero is a rest.
GATE	0–100% of the base step interval; zero is a rest.
LATE	-49% early to +49% late, using half-step look-ahead.
HARMONY	Adds a second note -12 to +12 semitones from the main note; zero disables it.

Up/Down and Down/Up omit repeated extremes; UpAndDown and DownAndUp repeat them. Fingered modes alternate the bottom or top note with the other held notes. Pitch-degree enables use a detected chord root, including common inversions. Root detection uses triad/seventh templates; ambiguous voicings may select a different root from another instrument.

Drag to draw; right-click or double-click for an exact value; wheel adjusts one unit. NOTE accepts ranks 1–8 or UP, DN, UD, DU, U+D, D+U, FLO, FHI and REST. The row below NOTE toggles mute without erasing values. Alt-click a NOTE step adds a motion-reset marker. Drag the top bar or use LEN to change length. LINK shares lengths; with LINK off each graph cycles independently. DIV slows the selected graph by 1–16 base steps. COPY / PASTE transfer a normalized shape, length and divider; RAND randomizes the selected graph. NOTE copying includes mute/reset flags. The pitch-class buttons cycle Pass, Block (x), Up (>) and Down (<); redirected notes move one semitone. These constraints

are global.

A–L are twelve named pattern memories. Switching slots automatically retains current edits. Click the name to rename; COPY TO duplicates the complete current pattern into another slot. RATE ranges from 1/512 note to 16 bars when SYNC is on, or 0.01–512 Hz otherwise. RETR resets motion with each incoming note. With RETR off and valid host PPQ, the clock aligns to transport. CHORD rotates the held voicing so the selected arp note is lowest, then plays the whole voicing. LATCH holds the released chord until a new chord arrives. Stopping the host or disabling the arp clears generated notes. Only one pattern plays at a time.

MIDI OUT sends generated notes and supported pitch/controller/pressure messages to the host. MIDI ONLY also silences synth output; turning it on enables MIDI OUT. Route output in the DAW or choose an output device in Standalone settings. Turning MIDI OUT off sends all-notes-off. MIDI routing depends on the host. Large SysEx messages are not forwarded. Early/late timing is sample scheduled; an early first note is clamped to the incoming note time, never before it.

Previous Pattern engine

Turn GRAPHS off to use the 0.4 Pattern grid, with note rank, octave, velocity, gate, repeat, chance and tie rows, independent lane lengths and two 16-step pages. Draw or right-click for values. Existing 0.4 projects select this engine automatically. Its pattern selector also uses the A–L memory bank. Legacy remains a separate mode with its original controls.

Legacy mode and compatibility

Legacy retains the ten original modes, velocity modes, 16 steps, transpose, wrap, gate, swing, octaves and hold. The Step sequence factory sound uses it. Older logisth states select Legacy automatically when they have no Pattern engine parameter. ARP enable clears held notes; play fresh notes afterward. Pattern uses note-rank/octave/gate lanes instead of Legacy transpose/wrap and its shared gate. Legacy controls remain available in the Legacy view.

7. Effects rack

Signal order is fixed: Distortion Phaser Chorus EQ Reverb Delay Compressor Main. Click a name to edit it and use the adjacent square to bypass or enable it. All seven effects are global after the A/B mix.

Section	Controls and use
Distortion	Overdrive, Foldback, Clip, Decimate and Bitcrush; Amount and Dry/Wet set character and blend.
Phaser	Six-stage stereo phasing with centre frequency, spread, L/R offset, width, LFO, feedback and Dry/Wet. Its frequency is a modulation destination.
Chorus	Delay, depth, rate, feedback, dual mode, width and Dry/Wet thicken a sound.
Equalizer	Bass and treble shelves with frequency controls and optional positive-gain compensation.
Reverb	Size, damping, predelay, width and Dry/Wet shape the room tail.
Delay	Independent left/right times or divisions, feedback, ping-pong, spread, width, low/high cut, smear and Dry/Wet.
Compressor	Stereo-linked threshold, ratio, attack and release control dynamics.

When SYNC is on, rate/time fields that support synchronization become division menus. Turning SYNC off exposes their independent Hz/time values again. Bypass can freeze a section's internal memory; a tail may resume when re-enabled. Delay memory is limited to 16 seconds per channel, including very slow synced divisions. Compressor attack/release and some effect delays use milliseconds; envelope times use seconds internally. The panel displays suitable units.

8. Mixer and performance

The mixer scope has A, B and L+R source buttons. A/B show each part's stereo sum after its filter, amplitude envelope and Mix level, before shared effects and Solo, scaled by MAIN. L+R shows the actual final left-plus-right output, after effects, MAIN and MIDI ONLY. It is a sum, not an average. The scope uses zero-crossing alignment and automatic visual scaling; the peak label gives the unscaled peak. Selecting a source changes only the view and is saved in state.

MIX A and MIX B set the part levels; MAIN controls output after all effects. Independent L/R meters show post-effects output. There is no hard output limiter: loud patches, resonance and effects may exceed 0 dBFS. Reduce MAIN as needed.

POLY sets up to 16 simultaneous notes. Each note may contain several unison voices. SOLO isolates A or B. MONO / LEGATO uses last-note priority and returns to the previous held note when the newest one is released. POLY is disabled in mono mode. GLIDE sets portamento time; Normal mode glides for overlapping notes, while Slide also permits glide between separate notes.

BEND / st sets pitch-wheel range in semitones. The on-screen BEND wheel springs back to the centre after dragging. MOD remains where you set it. Route Mod Wheel through a Misc block to hear its effect. MIDI pitch bend and modulation messages also update these controls. The keyboard covers five octaves.

9. Presets, automation and project recall

The ten factory sounds are Init / analog, Wide supersaw, Sub bass, Slow atmosphere, Pulse pluck, Step sequence, Morph glass, Sample keys, Grain cloud and Chord pulse. The sound selector lists factory sounds and user presets from the current folder. Its arrows step through that list. Loading replaces the current patch.

The PATCH menu provides Save as, Save current, Load file, Choose folder, Refresh library and Delete current. Save as creates a named **.logisth** file; Save current overwrites the active user preset. Delete asks for confirmation and moves the file to the OS trash. The default library is logisth/Presets under the OS application-data directory. Choose folder selects an instance's working library; it does not relocate files. Load file can open a preset from elsewhere. Save/load run on a worker, with completion/errors in the footer. Wait for the completion message before closing the instance or moving a newly saved file.

Presets contain all 936 audio parameters, routing, compressed audio for all four slots, WAV regions, chord mappings, twelve named patterns, theme and scope selection. The original sample files are unnecessary on recall. Saving uses a temporary file and atomic replacement. Invalid files leave the current patch intact. State/preset files are limited to 64 MiB. These are logisth's own files; renaming another format to .logisth does not convert it.

All 936 parameters are available for host automation. The first 514 parameter IDs and order from 0.4 are retained, with 422 IDs appended. A/B views, rack pages, theme and scope source do not change the patch's audio settings. DAW state contains the same embedded material and pattern data as a user preset. Restoring older states automatically selects the corresponding earlier granular/arp engines. The 0.4 destination expansion still affects normalized destination automation from earlier releases; inspect such automation when importing an older project.

There is no Sylenth1 bank import or MIDI Learn menu. Legacy prototype states receive a best-effort migration of filter and modulation settings; changes to DSP mean that identical sound is not guaranteed.

10. Sound-design examples

For a wide lead, load Wide supersaw. Raise unison detune gradually, open shared CUTOFF, and add a little Delay. Use MONO / LEGATO with a short GLIDE for connected melodies. Lower MAIN if the combined voices become too loud.

For an envelope sweep, load Init / analog, lower Filter A cutoff, select Cutoff A as Mod Env 1's first destination, and add a small positive amount. Set attack near zero, moderate decay, low sustain and short release. Increase the amount until the filter opens sufficiently at the start of each note.

For vibrato, choose Mod Wheel in a Misc source block and route it to LFO 1 Gain. Set LFO 1 to a sine wave in FREE mode at a few Hz, set its base Gain to zero, then route LFO 1 to Pitch A+B with a small amount. The wheel opens the vibrato depth. Keep amounts low to avoid large pitch excursions.

For a sequence, load Step sequence and hold a note. Adjust PITCH and VELOCITY, set a zero velocity for a rest, and enable HOLD to tie steps. Select 09–16 and increase LENGTH to use the second half. Automate the shared filter cutoff to move both parts together without replacing their local settings.

11. Troubleshooting and technical notes

High pitches in Sampler/Granular use interpolated resampling and may alias. Wavetable mip bands reduce aliasing, but phase warp can introduce new harmonics. These engines are independent implementations, not models of commercial synths. Dense granular patches with high unison/polyphony require more CPU. Imported material can be loud; use moderate levels. An import failure leaves the previous material intact. A source starts again on the next note or engine/material change.

Version 0.5 adds the documented Granulizer workflow and eight-graph arp. This is a development version. Native release archives are published only after their platform builds and tests pass. See BUILD_INFO.json in each archive for source revision and validation results.

No sound: check the audio/MIDI device or DAW routing, oscillator Voices and Volume, filter input, envelope sustain, Mix levels and MAIN. Check SOLO and whether the arp needs fresh notes. A long attack or low cutoff can hide a note.

No modulation: check the source is active, destination is not None, and amount is nonzero. The mod wheel needs an assigned route. A source can be zero during part of a note; Step Velocity needs an active arp gate.

Missing control: A/B selects the part, a rack name selects effect controls, and SYNC/FREE selects time versus rhythmic controls. RETRIG enables phase editing. The second group of arp steps is under 09–16.

High CPU: reduce unison voices, polyphony or active effects. Some abrupt changes to routing, waveforms and delay times can click. Selected continuous controls are smoothed, but not every transition is crossfaded.

Build validation includes audio at 44.1/48/96 kHz, filters, modulation, arp timing, state restoration, preset output and editor bindings/layout/theme persistence. Linux tests additionally load the actual CLAP and VST3 binaries. A VST3 factory smoke test does not replace testing in your particular DAW. Native macOS tests run on Intel; Universal architecture checks establish that arm64 code is present, not that it has been auditioned on Apple Silicon hardware.

The filter is an independent SVF design; reverb uses JUCE Reverb. Modulation scales and some sequencer behaviour are implementation choices. No sample-exact comparison with Sylenth1 has been performed. Third-party font notices are included in assets/fonts. The original Sylenth1 manual and graphics are not redistributed.